SDG 3: Good Health and Well-Being

AI-Based Fatigue Detection and Wellness Care System

Relationship: Primary and direct project alignment

Sustainable Development Goal 3 seeks to ensure healthy lives and promote well-being for people of all ages. The AI-Based Fatigue Detection and Wellness Care System, AURA, relates most directly to this goal because it addresses the challenges associated with prolonged screen-based activity. Students and professionals frequently work for extended periods without recognizing changes in comfort, concentration or posture. The project introduces a wellness-oriented approach that makes possible signs of fatigue more visible and encourages timely self-care. Its purpose is to support healthier digital routines rather than diagnose disease, confirm a medical condition or replace professional healthcare advice.

The system combines several observations instead of depending on a single indicator. Webcam-based facial analysis provides information about blinking, eyelid closure, mouth opening and head posture, while interaction tracking measures typing cadence, mouse movement and continuous screen time. These observations contribute to an estimated fatigue score. The personal work-pace component compares later interaction patterns with an individual's starting activity rather than assuming that every person should type or move a pointer equally quickly. This individualized comparison is useful because naturally slower interaction is not necessarily evidence of tiredness, and inactivity may simply represent reading or reflection.

AURA connects monitoring with short wellness interventions. Depending on the estimated condition and recommendation logic, the application can suggest eye relaxation, deliberate blinking, gentle stretching, posture correction, hydration or breathing exercises. These activities are intended to create opportunities for recovery without requiring users to abandon their work completely. Personalization can help make reminders more relevant than a fixed schedule that interrupts every user at the same interval. However, the system should not imply that a short activity can treat an underlying health problem or eliminate the need for adequate sleep and meaningful rest.

The Daily Wellness Dashboard extends the contribution beyond an individual warning. By displaying recorded fatigue estimates, screen time, blink trends and completed breaks, it gives users a way to review their digital habits over time. A user may identify long periods without breaks, compare different study routines or notice repeated discomfort during particular activities. Such information can support informed decisions about work duration, desk arrangement, lighting and rest opportunities. Dashboard values must be presented as estimates and observations, not as clinical measurements or proof that a person's health has improved.

Responsible implementation is essential to this goal. Camera frames are processed locally, and only the required numeric features are used for scoring. Users should understand what is being monitored, retain control over available settings and be able to pause monitoring. The application must also distinguish a missing face from a failed camera or stalled tracking engine. Displaying an unavailable estimate is more responsible than showing default values as if they were real observations. This helps avoid misleading reassurance, unnecessary warnings and excessive dependence on an automated score.

Overall, the project's strongest contribution to SDG 3 is the combination of awareness, personalized wellness support and opportunities for healthier screen-use behaviour. Whether these features actually reduce discomfort or improve well-being requires careful user evaluation, appropriate outcome measures and, where relevant, professional guidance. The project should therefore be described as a preventive wellness-support tool with potential benefits, rather than a medically validated fatigue detector. This distinction keeps the SDG discussion accurate while recognizing the practical value of encouraging regular self-care during digital work.

SDG 4: Quality Education

AI-Based Fatigue Detection and Wellness Care System

Relationship: Strong supporting alignment through digital learning

Sustainable Development Goal 4 promotes inclusive and equitable quality education and lifelong learning opportunities. AURA supports this goal through its relevance to the growing amount of learning conducted on digital devices. Students use computers for online classes, research, programming, examinations, assignments and collaborative projects. These activities can involve long sessions of close visual attention and limited movement. The project responds to this learning environment by encouraging students to recognize possible fatigue and manage their study routines more consciously. Its educational contribution concerns healthier participation and self-management, not a guarantee of higher marks or improved examination results.

A central feature is the use of individual observations rather than a uniform assumption about how long every student can study comfortably. Blink patterns, head posture, screen duration and interaction changes provide context for the estimated fatigue score. The starting-pace comparison is particularly relevant in education because students differ in typing experience, language familiarity and the type of assignment they are completing. A student learning to code may type slowly while thinking carefully, whereas another may type rapidly during note-taking. The system should interpret these differences cautiously and avoid equating speed with intelligence, effort or academic ability.

Personalized wellness recommendations can support a more balanced approach to digital study. A short visual break may be appropriate after sustained screen attention, while a posture reminder or gentle stretch may be more relevant during an extended writing task. Breathing exercises can provide a structured pause when students feel mentally overloaded. By offering brief interventions, the project aims to fit self-care into existing study activities rather than treating breaks as a failure to work hard. Students should nevertheless remain free to decide whether a reminder fits their current learning task and personal needs.

The dashboard offers a practical form of reflective learning about study habits. Students can review their screen-use patterns, recorded estimates and break completion rather than relying entirely on memory. This may help them plan shorter study blocks, arrange rest periods before lengthy assignments or recognize that repeatedly ignoring discomfort is not a sustainable strategy. The records can also support discussions about digital wellness in a classroom or mentoring session, provided participation is voluntary. Individual dashboards should not become an attendance mechanism, a disciplinary record or a basis for judging student commitment.

The project also has value as an educational example of interdisciplinary engineering. It connects computer vision, geometric feature extraction, machine-learning prediction, recommendation algorithms, database design and user-interface development within one application. Students building or reviewing it can learn about the difference between a model prediction and a reliable real-world conclusion. In particular, training with synthetic data does not establish performance on diverse students in ordinary classrooms. Explaining this limitation encourages responsible technical understanding, critical evaluation and honest reporting of what an academic prototype can and cannot demonstrate.

AURA therefore aligns with SDG 4 by supporting digital-wellness awareness and illustrating a practical application of responsible computing. Its potential educational benefits should be evaluated through usability studies, accessibility feedback and voluntary student participation. Different lighting conditions, camera quality, disabilities and learning activities may influence how useful the system is. With these limitations acknowledged, the project can contribute to a learning culture in which concentration, physical comfort, rest and personal choice are considered alongside academic productivity. Quality education is supported when technology helps learners participate sustainably rather than simply demanding longer periods of screen exposure.

SDG 8: Decent Work and Economic Growth

AI-Based Fatigue Detection and Wellness Care System

Relationship: Strong supporting alignment through workplace well-being

Sustainable Development Goal 8 includes productive employment, decent work and safe working environments. AURA connects with this goal through the realities of computer-intensive occupations. Software developers, designers, analysts, administrators and remote workers may spend much of their working day seated in front of a screen. Extended sessions can involve physical discomfort, reduced alertness and difficulty sustaining attention. The project introduces a way to make possible fatigue more visible and to encourage short opportunities for recovery. Its intended workplace contribution is healthier work practice, not an instruction to continue working when meaningful rest is needed.

The multimodal approach reflects the fact that work conditions cannot be understood from keyboard activity alone. AURA considers facial observations, posture-related indicators, screen duration and changes in interaction pace. Comparing an individual's current activity with their starting pace provides more relevant context than ranking workers against a single typing-speed standard. Even then, a slower pace may result from a complex task, a discussion, careful reading or a change in responsibility. The estimate must therefore be treated as an imperfect wellness signal rather than a measurement of employee productivity, motivation or competence.

Personalized interventions provide the connection between monitoring and practical workplace support. Instead of issuing identical messages at fixed intervals, the system can recommend activities such as eye relaxation, posture adjustment, hydration or a brief breathing exercise. These recommendations aim to create manageable recovery opportunities within a working day. The contextual bandit component uses feedback about interventions to help personalize later choices. Its learning concerns the recommendation process; it does not establish that a particular intervention has medically reduced fatigue. Workers should be able to decline, pause or postpone a recommendation according to their circumstances.

The dashboard can help an individual reflect on patterns across working sessions. For example, a user may notice that long uninterrupted tasks coincide with repeated warnings or that certain routines include very few breaks. This information can encourage discussion about workload, rest opportunities and workstation design. However, an application cannot correct unreasonable deadlines, excessive working hours or unsuitable workplace arrangements on its own. Employers remain responsible for appropriate working conditions. The existence of a wellness tool should never be used to transfer responsibility for organizational problems entirely onto individual workers.

Privacy and autonomy determine whether the project genuinely supports decent work. Appropriate use requires clear explanations, voluntary participation and limits on access to personal records. Camera frames should remain local, and the application should avoid collecting typed content or unnecessary information. Individual scores should not be used for hiring, promotion, disciplinary action or comparisons between colleagues. Any future organizational reporting would require a separate consent and governance process. These safeguards matter because a tool designed for employee care can become harmful if it is repurposed as a surveillance or performance-control mechanism.

The project's alignment with SDG 8 is therefore strongest when it supports the worker's own understanding and decision-making. Benefits such as fewer errors, greater productivity or reduced absence must not be claimed without evidence from suitable evaluation. A responsible assessment would examine usability, perceived usefulness, interruptions and privacy concerns alongside any measured outcomes. Used within these boundaries, AURA can demonstrate a worker-centred approach to digital wellness. It encourages sustainable participation in screen-based employment while recognizing that healthy work requires both individual support and responsible organizational practices.

SDG 9: Industry, Innovation and Infrastructure

AI-Based Fatigue Detection and Wellness Care System

Relationship: Strong supporting alignment through applied technological innovation

Sustainable Development Goal 9 promotes resilient infrastructure, inclusive and sustainable industrialization, and innovation. AURA relates most clearly to the innovation component of this goal. The project integrates computer vision, interaction analysis, machine-learning scoring, personalized recommendations and visual analytics to address a practical digital-wellness problem. Its academic value lies in showing how separate technologies can be organized into a coherent application rather than presenting an algorithm in isolation. This is a focused contribution to human-centred computing; it should not be described as evidence that the project has achieved broader national infrastructure or industrial-development objectives.

The perception layer demonstrates one part of this integration. Facial landmarks support the extraction of eye and mouth geometry, blink-related observations and head-posture information. Keyboard and pointer measurements add a different form of context about the user's work session. These signals are numeric summaries rather than a continuous upload of camera footage or typed text. The design illustrates how an ordinary computer can support an intelligent application without requiring a dedicated wearable device. Its feasibility still depends on a suitable browser, functioning webcam, adequate processing capability and practical environmental conditions.

The scoring and recommendation layers provide another aspect of innovation. The Python backend uses LightGBM to estimate a fatigue score from the available feature set, while a separate personal-pace adjustment represents sustained changes relative to a starting reference. A contextual bandit supports the selection of wellness interventions using feedback. These components demonstrate different forms of algorithmic reasoning within one workflow. The public static application uses a browser-based fallback rather than the hosted Python model, so reports and demonstrations should identify the actual scoring path being used instead of treating both implementations as equivalent.

A modular architecture makes the project useful for further development. The frontend, backend, perception logic, data storage and Guardian extension have distinct responsibilities. This separation allows developers to investigate a component without replacing the entire system. For example, tracking reliability can be improved independently of the sound-selection interface, while a future evaluation could compare alternative scoring approaches using the same recorded numeric inputs. Documentation, source-code organization and repeatable tests support this process. They make the prototype easier to inspect, reproduce and extend, although they do not by themselves prove real-world reliability.

Responsible innovation also requires attention to technical failure and limitations. A camera stream does not automatically mean facial analysis is functioning, and missing observations should not be interpreted as confirmed fatigue or absence. The project therefore distinguishes tracker readiness, valid detections and unavailable measurements. Similarly, synthetic training data can support development and testing but cannot substitute for representative evaluation with real users. Accuracy, fairness, responsiveness and recommendation usefulness must be investigated before wider deployment. A clear account of these uncertainties is part of the innovation process, not a weakness to conceal.

Through this perspective, AURA supports SDG 9 as an example of accessible, application-oriented engineering with potential for further research. Future improvements could address resource efficiency, accessibility, device compatibility and stronger validation. Any transition from a student prototype to an institutional or commercial product would also require appropriate security, maintenance and user-support arrangements. The project's contribution is therefore a practical foundation for responsible digital-wellness innovation. It demonstrates the design and integration of intelligent components while recognizing that successful infrastructure and trustworthy deployment require continued work beyond the initial implementation.

SDG 10: Reduced Inequalities

AI-Based Fatigue Detection and Wellness Care System

Relationship: Conditional supporting alignment through inclusive design

Sustainable Development Goal 10 seeks to reduce inequalities within and among countries. AURA has a supporting connection to this goal when it is designed to accommodate different users rather than assuming a single normal pattern of work. People vary in typing experience, reading habits, physical movement, vision, age, disability and familiarity with technology. A system that treats every difference as fatigue could disadvantage users whose interaction patterns do not match its assumptions. The project's contribution to this goal is therefore conditional: inclusive intentions must be accompanied by accessible design, appropriate controls and careful evaluation across different groups.

The personal starting-pace comparison provides one relevant design principle. Instead of concluding that a person is tired merely because their typing speed is below a population threshold, the system compares later active work with that person's own initial reference. Keyboard and mouse observations are considered separately so that one input does not have to be used continuously for the other to be meaningful. This can help avoid some misleading comparisons between users. Nevertheless, the starting reference may already reflect tiredness or an unsuitable task, and individual baselines do not automatically remove bias or establish fairness.

Flexible input preferences are also important. Some people work mainly with a keyboard, use assistive input devices, rely on touch interaction or read for long periods without moving a pointer. Others may not be able or willing to use a camera. An inclusive wellness application should explain which measurements are available, permit users to disable appropriate inputs and avoid treating missing activity as evidence of poor performance. Alternative access methods and meaningful use without a webcam would require additional design and evaluation. These should be identified as improvement opportunities rather than presented as fully implemented capabilities.

Using standard computer hardware can reduce the need to purchase a specialized wearable device, which may lower one barrier to trying the application. However, this does not make the system universally accessible or free from cost. Users still need a suitable device, browser and working environment, and some features depend on camera quality or sufficient computing resources. Internet access may also be needed to obtain and update the application. An honest SDG discussion must recognize these remaining barriers. Accessibility cannot be inferred solely from the absence of a dedicated sensor or from the publication of source code.

Computer-vision performance requires particular attention. Lighting conditions, spectacle glare, camera position and other visual differences can affect whether reliable landmarks are obtained. Before making claims about equal performance, the system would need representative testing and an analysis of failure rates across relevant conditions and user groups. AURA's ability to report unavailable tracking is useful because a technical failure should not be presented as a person's fatigue state. Clear explanations, readable controls, keyboard navigation and accessible notifications are additional priorities, but their effectiveness should be checked through an accessibility review and direct user feedback.

Accordingly, AURA's relationship with SDG 10 is best described as a commitment to more individualized and inclusive wellness support, not demonstrated elimination of inequality. Future development should involve diverse participants and consider how benefits and limitations are distributed. The application must not be used to rank students or workers against each other or disadvantage those with different abilities. With these conditions respected, personal baselines, privacy controls and transparent handling of missing measurements can contribute to a more equitable design approach. The central principle is that technology should adapt to people rather than require every person to fit one behavioural standard.

SDG 12: Responsible Consumption and Production

AI-Based Fatigue Detection and Wellness Care System

Relationship: Indirect and limited alignment through responsible digital design

Sustainable Development Goal 12 promotes responsible consumption and production. AURA has an indirect relationship with this goal through the way a digital-wellness application can be designed, maintained and evaluated. Its core purpose is user well-being, not environmental monitoring, energy management or waste reduction. Therefore, the project should not claim that issuing break reminders automatically reduces electricity use or carbon emissions. A defensible connection focuses instead on avoiding unnecessary dedicated equipment, limiting unnecessary data collection and encouraging thoughtful use of existing computing resources. Any environmental benefit remains a possibility that would need separate measurement.

The use of an ordinary webcam and existing keyboard and mouse provides one practical consideration. The project does not require a special wearable device solely to collect the selected observations. Where suitable hardware is already available, this design may avoid an additional equipment purchase for the prototype. It demonstrates an approach in which software adds a function to an existing device rather than always requiring a new product. However, the project has not measured avoided material use or electronic waste. It would be inappropriate to turn this design choice into a quantified environmental claim without an appropriate comparison.

Data handling is another area where responsible digital design matters. Processing camera frames locally and transmitting only required numeric features limits the need to move and store continuous video for this workflow. Avoiding typed-content collection also reduces unnecessary information handling. These choices are relevant to privacy and proportionate system design, but they do not by themselves prove a lower overall environmental footprint. Local inference consumes processing power, and the application may use resources that would otherwise remain idle. A fair assessment would consider both reduced data movement and the additional computation introduced by the monitoring process.

Software maintenance can also support a resource-conscious approach. A modular project allows a developer to repair a component, update an interface or improve a model-loading process without redesigning the entire application. Clear documentation and source availability can assist reuse and reduce avoidable duplication of development effort. Compatibility with existing devices is another useful objective because unnecessary hardware replacement should not be treated as the default solution to a software problem. These are design principles rather than proof that this particular project has extended a device's usable life or reduced waste in practice.

The project's wellness recommendations must also be described accurately in this context. A short break may encourage someone to step away from a screen, but the computer can continue consuming energy during that time. AURA does not currently establish verified energy savings, provide an environmental dashboard or automatically manage device power as part of the stated wellness workflow. Future research could examine processing load, inference frequency, storage requirements and safe power-management options. Such improvements should be evaluated against tracking reliability and accessibility so that resource reductions do not create misleading measurements or exclude users with less capable devices.

The SDG 12 connection is consequently secondary to the project's stronger health and educational links. Its value lies in encouraging developers to consider the material, computational and data costs of a useful application rather than assuming that software is resource-free. Responsible reporting should separate implemented choices from possible future improvements and measured results. By reusing available equipment, limiting unnecessary data and maintaining a repairable design, AURA can illustrate relevant principles of responsible digital development. Claims about actual reductions in energy, emissions or electronic waste should be reserved for evidence produced by a dedicated assessment.

SDG 17: Partnerships for the Goals

AI-Based Fatigue Detection and Wellness Care System

Relationship: Future enabling alignment through multidisciplinary collaboration

Sustainable Development Goal 17 emphasizes partnerships and cooperation in support of sustainable development. For AURA, this goal represents a pathway for future development rather than an already achieved project outcome. Digital fatigue involves technical, physical, psychological and organizational factors, so a responsible system cannot be evaluated through software implementation alone. Collaboration with relevant academic, professional and user communities could help improve the application and clarify its appropriate role. The project should not claim formal partnerships, institutional endorsement or shared research results unless such arrangements have actually been established and documented.

Educational institutions could provide one setting for carefully organized evaluation. Students, teachers and project supervisors may help identify whether the interface is understandable, whether reminders fit ordinary learning activities and which features create unnecessary interruptions. Participation would need to be voluntary, and students should not face academic consequences for declining monitoring or stopping a study session. Appropriate approval and consent procedures would be required before conducting research involving people or collecting identifiable information. This type of collaboration could connect engineering development with practical educational needs while protecting the interests of participants.

Health, ergonomics and human-computer interaction specialists could contribute different forms of expertise. They could review intervention wording, advise on appropriate evaluation measures and help distinguish wellness support from medical assessment. Their involvement could also help identify when an automated warning is insufficient and when users should be encouraged to seek meaningful rest or professional advice. A consultation would not automatically validate the fatigue score or authorize clinical claims. Any conclusion about health benefits, detection accuracy or intervention effectiveness would still depend on the design, quality and results of a suitable evaluation.

Technical collaboration could strengthen reliability and reproducibility. Developers working on computer vision, machine learning, accessibility and web security could inspect different parts of the system and propose improvements. A shared, documented feature schema and repeatable tests make it easier to compare implementations and investigate failures. Future collaborators might examine alternative models, more representative data or resource-efficient inference. Where human data is involved, access permissions, privacy safeguards and permitted uses must be agreed in advance. Synthetic development data should remain clearly identified rather than being presented as evidence obtained from a clinical or workplace population.

Workplace and accessibility organizations could help evaluate real use without turning the application into a surveillance tool. Such involvement could reveal differences between office tasks, assistive technologies, remote work and classroom conditions that a small development team may overlook. Agreements would need to define who controls data, whether information is shared, how long records are retained and how participants can withdraw. Broader reporting should avoid exposing individual behaviour or enabling employee ranking. Cooperation supports the project's goals only when it preserves user autonomy and respects the limits of the underlying measurements.

AURA therefore connects with SDG 17 through the opportunity to develop a responsible collaboration model around an existing academic prototype. Its documentation, source code and clear statement of limitations can provide a starting point for discussion, not evidence that a partnership already exists. Future work should specify the participants, responsibilities, consent arrangements and intended evaluation outcomes before making partnership claims. With these safeguards, cooperation could improve usability, accessibility and scientific credibility while connecting technological development to genuine user needs. The goal is not simply to add more organizations to a project description, but to establish accountable relationships that make the resulting wellness support more useful and trustworthy.